PASSWORD STRENGTH CHECKER USING PYTHON

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ABSTRACT

This project is developed using Python programming language to check the strength of a password entered by the user. The system analyzes the password based on several security conditions such as password length, uppercase letters, lowercase letters, numbers, and special characters. It also checks whether the password belongs to a list of commonly used passwords. Finally, the program categorizes the password as Weak, Medium, or Strong and generates a strong password suggestion for the user.

INTRODUCTION

Passwords are important for protecting personal and organizational data. Weak passwords can easily be hacked using brute force or dictionary attacks. Therefore, strong passwords are necessary for improving cybersecurity. This project helps users evaluate the quality of their passwords and encourages the use of stronger passwords.

OBJECTIVE

• To check the strength of a password
• To identify common weak passwords
• To encourage secure password practices
• To generate a strong password suggestion

MODULES USED

1. re Module
Used for regular expression matching.

2. random Module
Used to generate random passwords.

3. string Module
Used to access letters, digits, and special characters.

WORKING PRINCIPLE

1. The user enters a password.
2. The program checks password length, uppercase letters, lowercase letters, numbers, and special characters.
3. A score is calculated based on these conditions.

4. If the password matches common weak passwords, the score is reduced.
5. The password is classified as Weak, Medium, or Strong.
6. Finally, a strong random password suggestion is generated.

ADVANTAGES

• Easy to use
• Improves password security
• Detects weak passwords
• Generates secure passwords
• Beginner-friendly Python project

LIMITATIONS

• Checks only basic password conditions
• Does not store passwords securely
• No database integration

FUTURE ENHANCEMENTS

• Add GUI using Tkinter
• Encrypt passwords
• Store password history
• Add OTP authentication
• Create web-based version

CONCLUSION

The Password Strength Checker project helps users understand the importance of secure passwords. The project successfully checks password strength and generates strong password suggestions. This project is useful for learning Python basics, regular expressions, and cybersecurity concepts.

REFERENCES

1. Python Official Documentation
2. W3Schools Python Tutorials
3. GeeksforGeeks Python Security Articles

SOFTWARE REQUIREMENTS

Software	Version
Python	3.x
IDE	VS Code / PyCharm / IDLE

SOURCE CODE

```
import re
import random
import string

common_passwords = ["123456", "password", "admin", "qwerty"]

password = input("Enter Password: ")

score = 0

if len(password) >= 8:
    score += 1

if re.search(r"[A-Z]", password):
    score += 1

if re.search(r"[a-z]", password):
    score += 1

if re.search(r"[0-9]", password):
    score += 1

if re.search(r"[!@#$%^&*]", password):
    score += 1

if password in common_passwords:
    print("Common password detected")
    score -= 1

if score <= 2:
    print("Weak Password")

elif score <= 4:
    print("Medium Password")

else:
    print("Strong Password")

chars = string.ascii_letters + string.digits + "!@#$%^&*"

suggestion = ''.join(random.choice(chars) for i in range(14))

print("Suggested Strong Password:", suggestion)
```

SAMPLE OUTPUT

Example 1
Enter Password: hello123
Medium Password
Suggested Strong Password: A@9kLm#12XpQ$

Example 2
Enter Password: Admin@123
Strong Password
Suggested Strong Password: T#8mPq@5Xz!2Ab